SECTION A (25 marks)
This section consists of **3** questions. Answer **all** questions.

1. Evaluate each of the following limits, if exists.

   a) $\lim\limits_{x\to 3}\dfrac{2x-6}{-x^2+2x+3}$ [3 marks]

   b) $\lim\limits_{x\to 9}\dfrac{x-9}{3\left(\sqrt{x}-3\right)}$ [3 marks]

   c) $\lim\limits_{x\to -\infty}\dfrac{x+2}{\sqrt{4x^2+1}}$ [3 marks]

2. Differentiate the following with respect to x. Give your answer in the simplest form.

   a) $y=\tan^3\left(e^x+3\right)$ [3 marks]

   b) $y=\ln\left(\dfrac{2-x^2}{x(1-x)}\right)$ [4 marks]

   c) $y=2^{x^3\ln x}$ [4 marks]

3. The curve $y = 2x^3 + bx^2 + cx$ has a turning point at (1, 5).

   a) Find the values of b and c. [3 marks]

   b) By using the second derivative test, determine whether the stationary point at (1, 5) is a minimum or maximum point.

   [2 marks]

SECTION B (75 marks)
This section consists of **7** questions. Answer **all** questions.

1. a) Given $x = 3 + 2i$, $y = -i$ and $\dfrac{z}{13} + x = \dfrac{1}{x} + y^2$.
Express z in polar form. [6 marks]

b) Given that the complex number, express $z = \sqrt{-1}\left(2i - i^{10}\right)$ in Cartesian form. [3 marks]

2. a) Determine the possible values of x if $\left|\dfrac{2x}{3+x} + 1\right| \geq 2$.
Express your answer in interval form. [6 marks]

b) Find the value of x such that $2|x| \geq 3x - 10$. [5 marks]

3. Given $f(x) = 2x + 1$ and $g(x) = \dfrac{-3x}{x+1}$.

a) State the domain of $f(x)$ and $g(x)$. [2 marks]

b) Obtain $fg(k)$ and $gf(k)$. Hence, determine the constant value of k where $f \circ g(k) = g \circ f(k)$.
[8 marks]

4. Given

$$f(x) = \begin{cases} -|x+2|, & -5 \leq x \leq -1 \\ 3, & -1 < x < 1 \\ (x-2)^2 + 1, & x \geq 1 \end{cases}$$

a) Evaluate $f(-3)$. [1 marks]

b) Sketch the graph of $f(x)$. [3 marks]

c) Based on the graph, determine the point of x – intercept and y – intercept.
[2 marks]

d) Hence, find the domain and range of f. [2 marks]

5. Given

$$f(x)=\begin{cases} ax+7 \ , \ x<-k \\ 3 \ , \ -k\le x<1 \\ x^2+1 \ , \ 1\le x<k \\ 3x+1 \ , \ k<x<5 \\ 16 \ , \ x\ge 5 \end{cases}$$

a) Find the values of k if $\lim_{x\to k} f(x)$ exist. [5 marks]

b) Hence, by using the value of $k>0$, find the value of a if $\lim_{x\to -k} f(x)$ exist. [5 marks]

6. a) If $xy=2(x-y)^2$, find the following values at $\left(1,\frac{1}{2}\right)$

i) $\frac{dy}{dx}$ ii) $\frac{d^2y}{dx^2}$ [9 marks]

b) Given $x=\cos 2t$ and $x=2t+\sin^2 t$, show that $\frac{dy}{dx}=-\cos ec2t-\frac{1}{2}$ and evaluate $\frac{d^2y}{dx^2}$ when $t=\frac{\pi}{8}$. [7 marks]

7. a) Given a water tank which has a shape of circular cone with a base radius of 3 metres and a height of 5 metres. If water being pumped into the tank at the rate of 2m^3/minute, find the rate of change in water level rising when the water is 3 metres deep.

[5 marks]

b) A closed cylinder of height *h* cm and radius *r* cm. The volume of the cylinder is 128π cm^3, and its total surface area is A cm^2

i) Show that $A=2\pi\left(r^2+\frac{128}{r}\right)$. [3 marks]

ii) Calculate the minimum value of *A*. [3 marks]

END OF QUESTION

Final Answer:

SECTION A

1. $a)\ -\frac{1}{2}$ $b)\ 2$ $c)\ -\frac{1}{2}$

2. $a)\ 3e^{x}\tan^{2}\left(e^{x}+3\right)\sec^{2}\left(e^{x}+3\right)$ $b)\ -\frac{2x}{2-x^{2}}-\frac{1}{x}+\frac{1}{1-x}$

   $c)\ \left(1+3\ln x\right)x^{2}2^{x^{3}\ln x}\ln 2$

3. b = - 9 c = 12 b) maximum point

SECTION B

1. a) $z=7\sqrt{65}\left[\cos\left(-2.622\right)+i\sin\left(-2.622\right)\right]$ b) $z=-2+i$
2. a) $\left(-\infty,-3\right)\cup\left(-3,-\frac{9}{5}\right]\cup\left[3,+\infty\right)$ b) $\left\{x:x\leq 10\right\}$
3. a) $D_f=\left(-\infty,\infty\right);\ D_g=\left(-\infty,-1\right)\cup\left(-1,\infty\right)$

   b) $fg(k)=\frac{1-5k}{k+1},\ gf(k)=\frac{-6k-3}{2k+2},\ k=\frac{5}{4}$
4. a) $f\left(-3\right)=-1$ b) D.I.Y

   c) x – intercept $\left(-2,0\right)$, y – intercept $\left(0,3\right)$ d) $D_f=\left[-5,\infty\right);\ R_f=\left[-3,0\right]\cup\left[1,\infty\right)$
5. a) k = 0, k = 3 b) $a=\frac{4}{3}$
6. a) i) $\frac{1}{2}$ ii) 0 b) -2
7. a) $\frac{dh}{dt}=\frac{50}{81\pi}m/\min$ b) $A=96\pi\,cm^{2}$